Circuit investigation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use measurements to safely test a predicted circuit result.

SUCCESS CRITERIA - I can...

1. Draw and predict before powering a circuit.
2. Place ammeters in series and voltmeters in parallel.
3. Compare measured and predicted values using specific numbers.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “ammeter” and “voltmeter” to explain your answer. Sentence starters are optional.

ammeter: an instrument placed in series to measure current

voltmeter: an instrument placed across two points to measure voltage

trial: one run or repetition of a test

01 Warm-up: bring back an earlier idea

A circuit has **3** identical series resistors.
1. **Add two more** at unchanged ideal supply voltage.
2. **Compare new current** with the original.

Word help: current = rate of charge flow

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

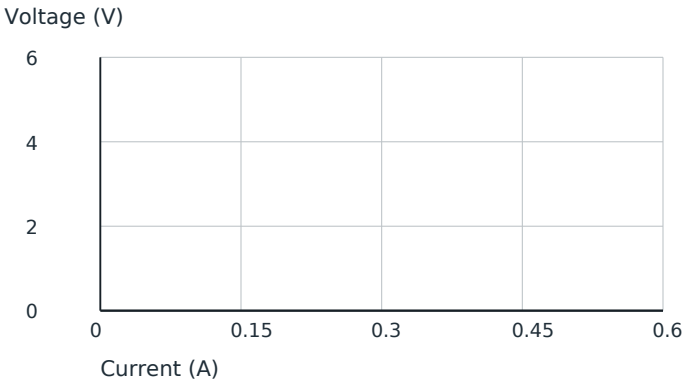
02 Worked example: explain the prediction

Reference: V = IR.
Use approved low voltage. Ammeter in series; voltmeter across a component. Disconnect before rewiring; never put an ammeter across the supply.

Worked reasoning: Supplied measurements show V=**2**, **4**, **6 V** and I=**0.2**, **0.4**, **0.6 A**. Each V/I is **10 ohms**. This supports treating the resistance as constant over the measured range.

Your turn: Plot V against I and **mark the slope**. **State why the data** do not show that the same relationship holds at every possible temperature or voltage.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

On a resistor circuit sketch, add an ammeter to **measure loop current** and a voltmeter to **measure the resistor’s drop**.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Apply the physics

A lab trial records **5.8 V** across a nominal **12 Ω** resistor and **0.46 A** through it.
1. **Calculate measured resistance V**/I and percent deviation from **12 Ω**.
2. **Use the table** to separate prediction, measurement and one possible source of measurement uncertainty.

Sentence starter: I predict ... . I used ... to make this prediction because ... .

Nominal R (Ω)	Measured V (V)	Measured I (A)	V/I (Ω)
12	5.8	0.46	___

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A reading disagrees with your prediction. **List two checks before changing** your explanation or deleting measurements.

Sentence starter: I predict … . I used … to make this prediction because … .

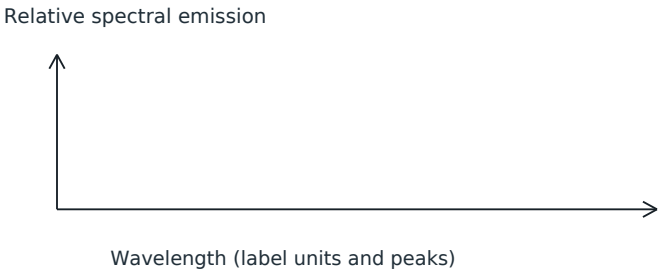
Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

An ideal blackbody has peak wavelength **580 nm**. Use $\lambda_{max}\,T$=**2.90×10⁶ nm·K**. **Estimate temperature**. **Show how the thermal peak** changes.

Word help: blackbody = an ideal absorber/emitter used in a thermal-radiation model

Sentence starter: I found … [unit]. This value means … in this situation.

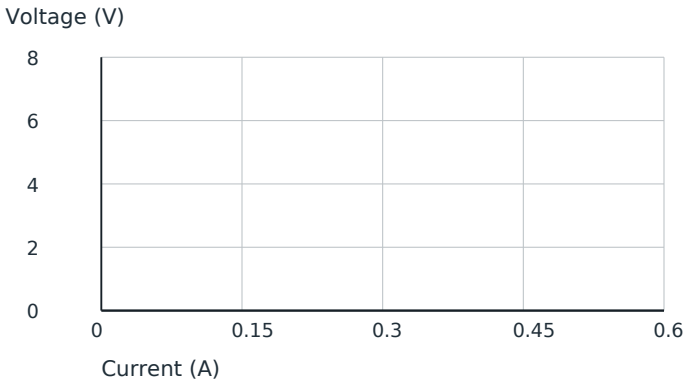


Marzano check - circle: ① ② ③

07 Exit check

1. Design a three-point V-I test.
2. **State what** to control and **what pattern** would support constant resistance.

Sentence starter: My design includes … . It does what is needed by … .



Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Two readings are **3.0±0.1 V** and **0.30±0.02 A**. **Find central resistance** and worst-case bounds.

Sentence starter: I found … [unit]. This value means … in this situation.

Quantity	Lower bound	Central value	Upper bound
Voltage (V)	____	____	____
Current (A)	____	____	____
Resistance (Ω)	____	____	____

Marzano check - circle: ① ② ③